Yingxin Guo

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About Me

I am a second-year master student at Shanghai Jiao Tong University, advised by Prof. Tong Ye. My research interests focus on optimizing optical networks for datacenter and AI computing networks.

Education

Shanghai Jiao Tong University *Sep. 2024 - Expected Mar. 2027*
Master of Information and Communication Engineering *Shanghai, China*

- GPA: 3.84/4.0
- Award: National Scholarship (2025)

Shanghai Jiao Tong University *Sep. 2020 - Jun. 2024*
Bachelor of Information Engineering *Shanghai, China*

- GPA: 3.88/4.3
- Award: Outstanding Graduates of Shanghai Jiao Tong University (2024)

Publications

- **Yingxin Guo** and Tong Ye, "OCS-based Double Resource Pooling for Flexible Intra- and Inter-rail Connectivity in AIDC Networks" in *Proc. Optical Fiber Communication Conference (OFC)*, 2026. (accepted)
- **Yingxin Guo**, Tong Ye, and Yaxuan Ma, "Second-Moment Vliant Load Balancing (sVLB) in Optical Data Center Networks," *IEEE Journal on Selected Areas in Communications (JSAC)*, vol. 43, no. 5, pp. 1793-1808, May. 2025.
- Yaxuan Ma, **Yingxin Guo**, and Tong Ye,"Opportunistic Load Balancing in Optical Datacenter Networks using Spare Capacity" in *Proc. IEEE OptoElectronics and Communications Conference (OECC)*, Jul. 2023.

Research and Project Experience

Optical Data Center Network Architecture Design *Sep. 2024 - Present*
OCS-Based Double Resource Pooling (ocs-DRP) Network for AI Data Centers *Shanghai Jiao Tong University*

- Propose ocs-DRP, an optical circuit switch-based flat and scalable AIDC network, supporting intra- and inter-rail connectivity for AI demand through double resource pooling.
- By simulation on SimAI platform, ocs-DRP achieves 40% lower power with latency comparable to an rail-optimized fat-tree.

Traffic Scheduling in Optical Data Center Networks *Jul. 2023 - Sep. 2024*
Second-Moment Vliant Load Balancing (sVLB) in Optical Data Center Networks *Shanghai Jiao Tong University*

- Propose a second-moment VLB (sVLB) framework for optical data center networks to perform load balance while suppressing residual traffic burstiness.
- Compared to existing strategies, sVLB has better delay performance and can achieve more than 50% reduction in the 99th percentile value of FCT slowdown.

Optical Data Center Networks Simulation Platform Development *Jan. 2022 - Jul. 2023*
Network Simulation Platform Based on Netbench *Shanghai Jiao Tong University*

- Develop a simulation platform for optical data center networks based on Netbench, supporting network topology and traffic generation, and multiple traffic scheduling algorithms.

Technical Skills

Programming Languages	Python, C++, Matlab, Java
Tools & Documentation	Git, Linux, LaTeX, Typst, OriginLab, MS Office
Language Skills	Chinese: Native, English: CET-6